Aror University of Art, Architecture, Design & Heritage Sukkur.

Department of Artificial Intelligence and Multimedia Gaming

Programming for AI (Spring-2026)

LAB No. 04

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Objective of Lab No. 04:

After performing lab 4, students will be able to:

- Implement functions in Python
- Return Multiple Values from Functions
- Use Different Type of Arguments in Functions
- Implement Lambda Function
- Use Map and Filter in Python

Lab Tasks:

Task#01: You are developing a Payroll Processing Module for a company.

The HR department wants a function that:

1. Accepts employee salary details.
2. Calculates:
   a. Gross Salary
   b. Tax deduction
   c. Insurance deduction
   d. Net Salary

e. Annual Salary

3. Returns multiple computed results. (gross, tax, insurance, net_salary, annual_salary) in a proper format as salary slip

Parameters:

4. `basic_salary` → Monthly base salary
5. `allowances` → Monthly allowances
6. `bonus` → Monthly bonus
7. `tax_rate` → Tax percentage (e.g., 12 for 12%)
8. `insurance_rate` → Insurance percentage (e.g., 5 for 5%)

Task#02 You are building an AI Research Evaluation Tool.

In real AI projects, we evaluate models using metrics like:

- Accuracy
- Precision
- Recall
- F1-Score
- AUC
- Loss

Different experiments may produce **different numbers of evaluation reports**.

So instead of fixed parameters, your function must:

- Accept multiple evaluation reports where each report is a **dictionary**

{

"model_name": "Decision Tree",

"accuracy": 0.91,

"precision": 0.88,

"recall": 0.86,

"f1_score": 0.87,

"loss": 0.25

}

It means your are going to use ***args.**

Calculate:

- Average Accuracy
- Best Performing Model (based on F1-score)
- Worst Performing Model (based on Loss)

Task#03: You are building a small preprocessing module for an AI model.

You are given a dataset of prediction probabilities from a binary classifier (values between 0 and 1).

Before feeding data into evaluation:

1. Remove invalid probabilities
2. Convert probabilities to percentage
3. Identify high-confidence predictions

probabilities = [0.95, 0.82, -0.4, 1.2, 0.67, 0.45, 0.99, 0.30]

Step 1 – Remove Invalid Values

Using filter() and lambda:

- Keep only values between **0 and 1**

Step 2– Convert to Percentage

Using `map()` and `lambda`:

- Convert each value into percentage

Step 3- Extract High Confidence Predictions

Using filter() and lambda:

- Select values ≥ 80%

Step 4 – Display

- Cleaned probabilities
- Percentage values
- High-confidence predictions